RACK & POWER



Simplify High-Density Deployments:
Reducing AC Whip Complexity in Multi-Power Shelves Open Rack

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OCP PLATINUM® Member

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Abstract:

As high-density AI workloads drive ORV3 adoption, powering multiple 33kW/72kW power shelves requires excessive power whips, causing cabling complexity and brownfield AC incompatibilities. This session introduces an open, modular AC PDU supporting multiple independent modules with blade-level short-circuit protection, optimizing the layer between AC facility and the ORV/ORW integrated racks. Accelerating open standards requires solving real-world deployment friction to a wide range of customers. By using this AC PDU for power shelves, a reduction of power whips per rack can be achieved which resolves space constraints and input mismatches without cost-prohibitive facility overhauls. It provides enterprise and hyperscale operators a practical bridge to ORV3 ecosystems, advancing OCP tenets of compatibility, efficiency, and scalability.

Social summary:

Deploying ORV3 Racks with multiple power shelves? Learn how a new modular, AC PDU for Power Shelves simplifies cable management, solves AC input incompatibilities, and includes blade-level protection to accelerate OCP adoption. Join us at the #OCPSummit2026!

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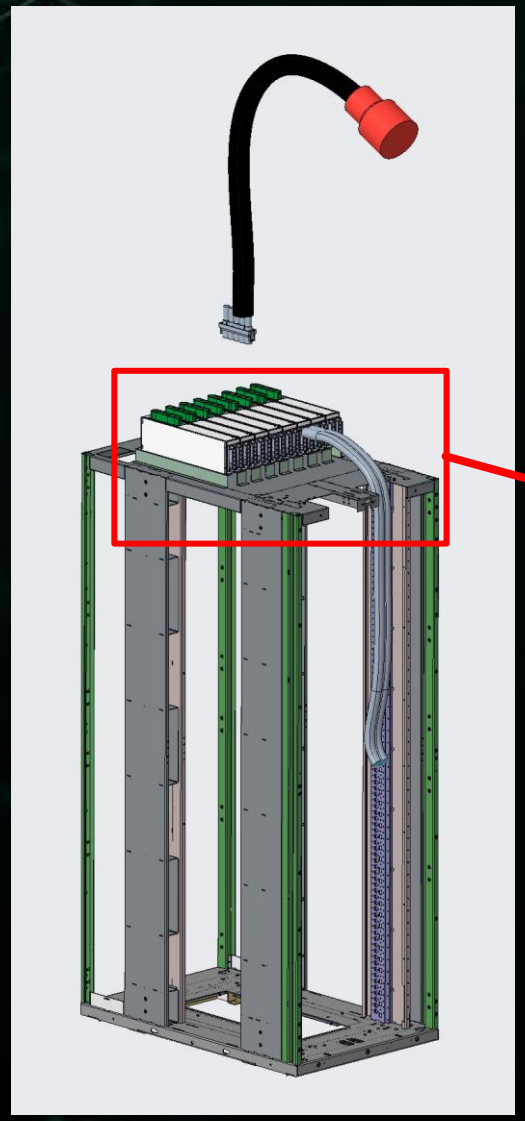
Proposal # 1
Horizontal AC High Power PDU on Top of Rack

Proposal # 2
Vertical High Power PDU hang on Rear of Rack

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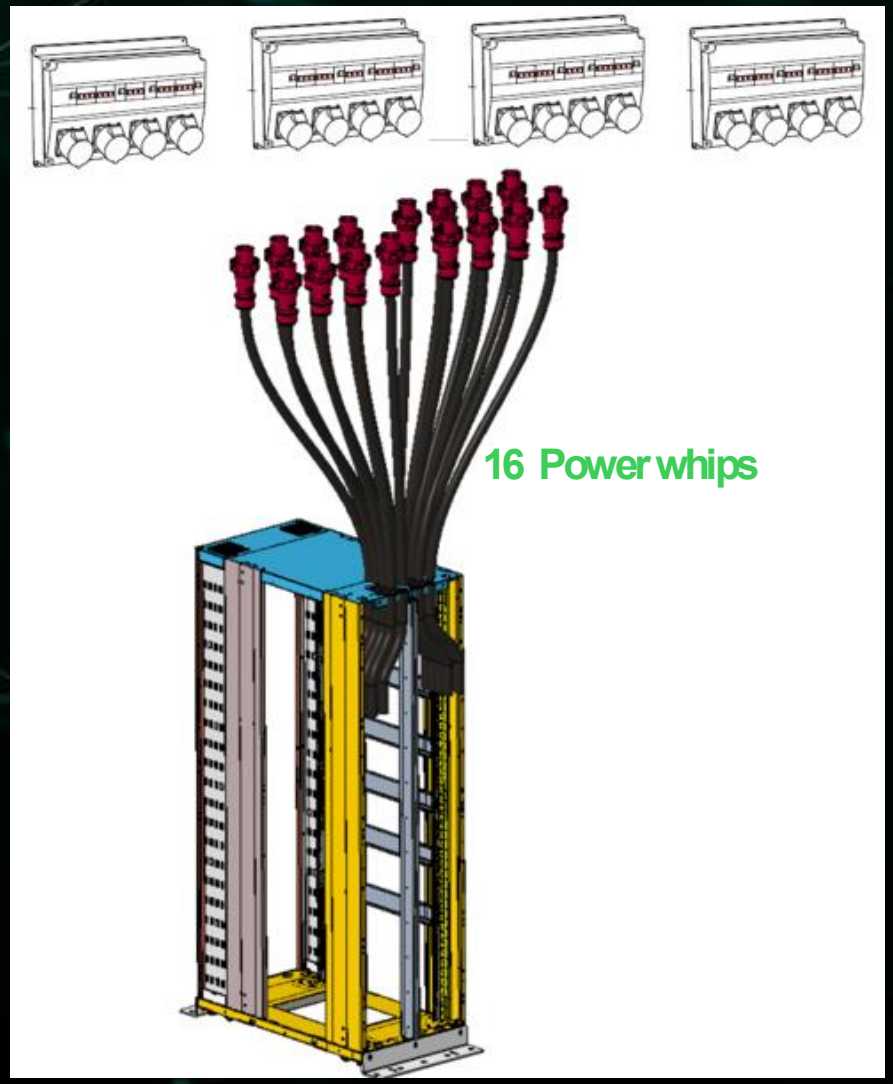


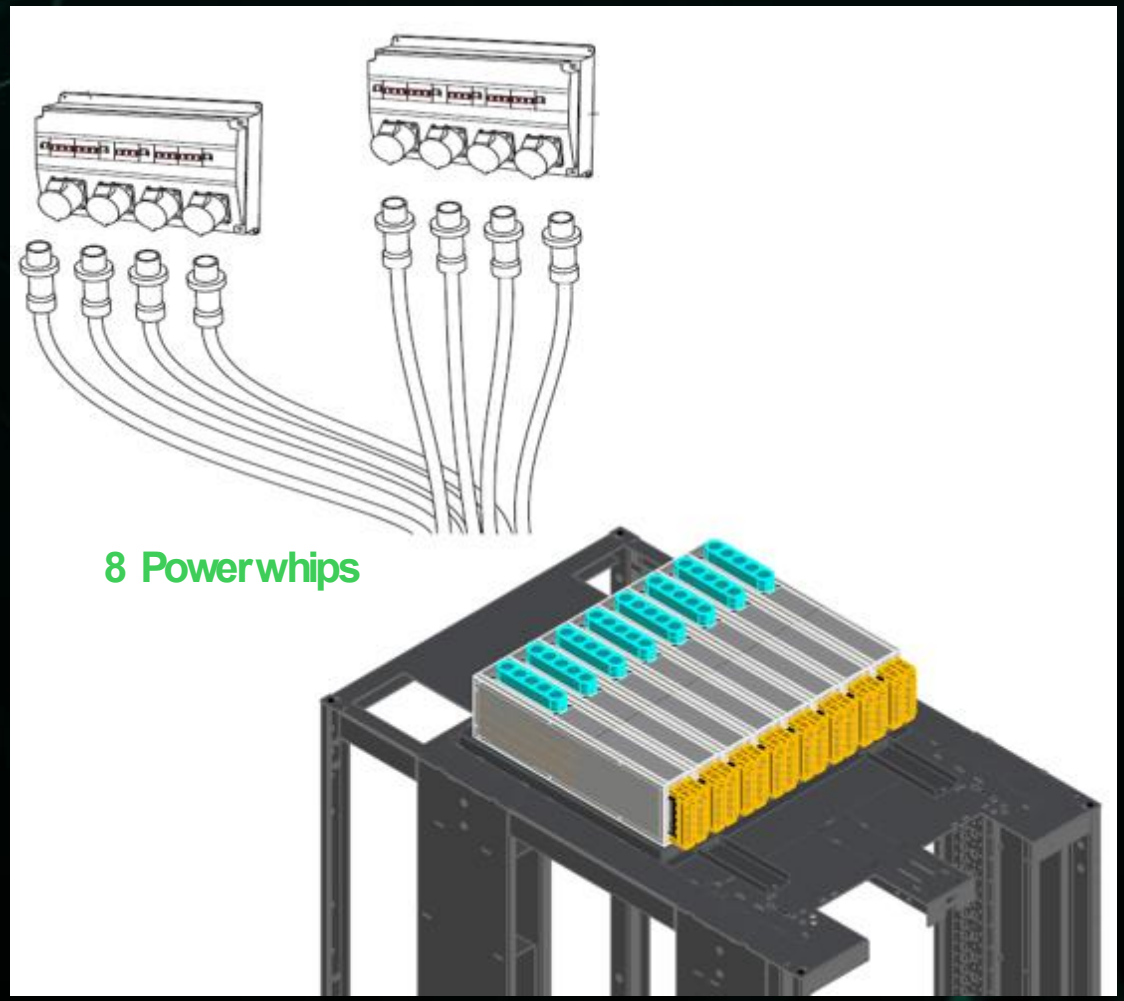
From Bus or RPP
125A

Feeds
PowerShelf
(2x) 60A

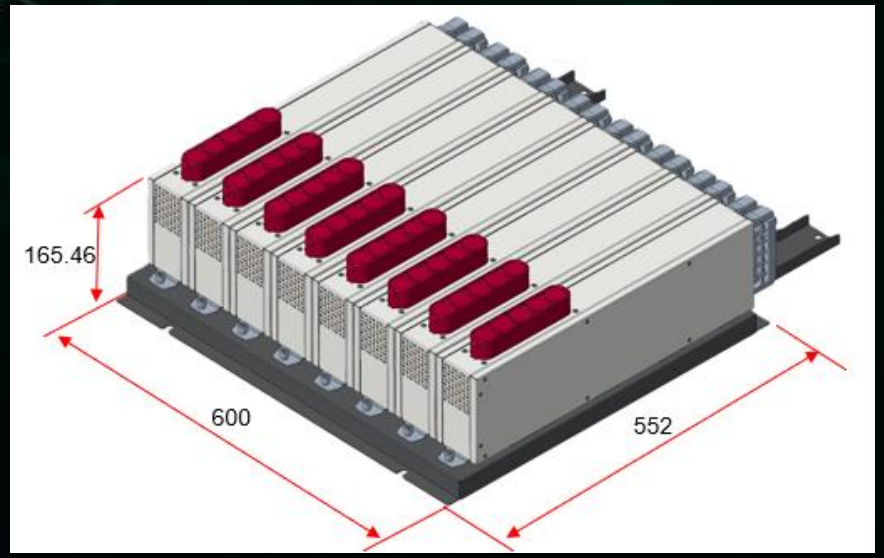
8 Feeder Box
Modules

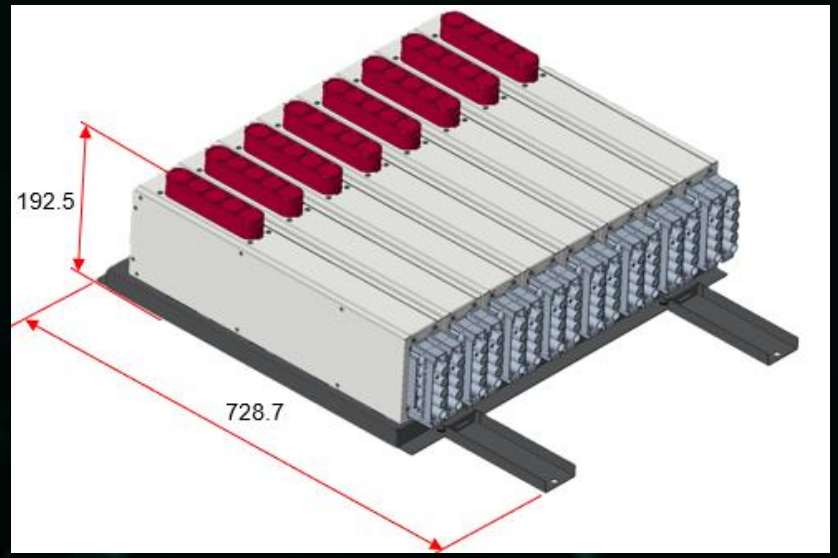
Maximum Reducing AC Whip From 16 to 8





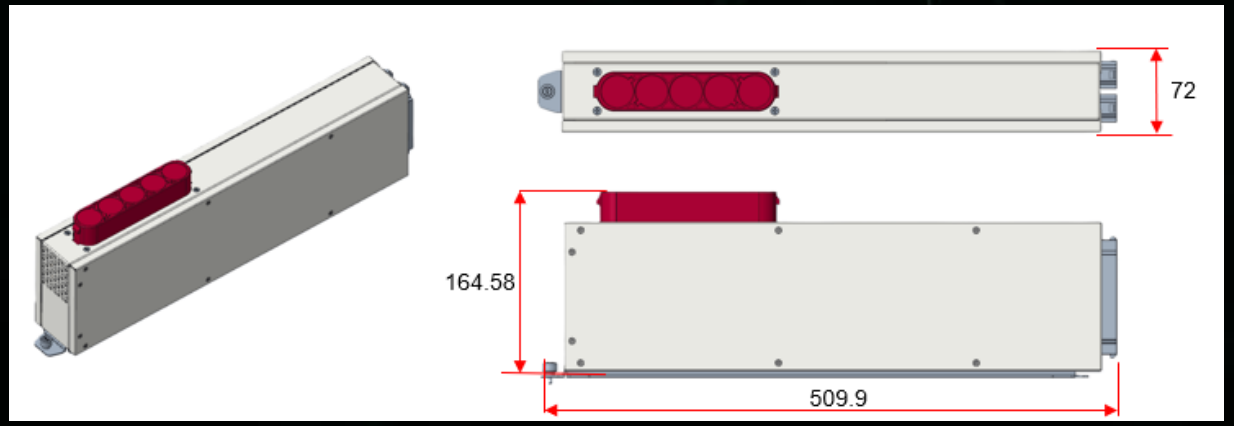
Dimensional drawings of the whole solution, the base and Feeder Box modules.





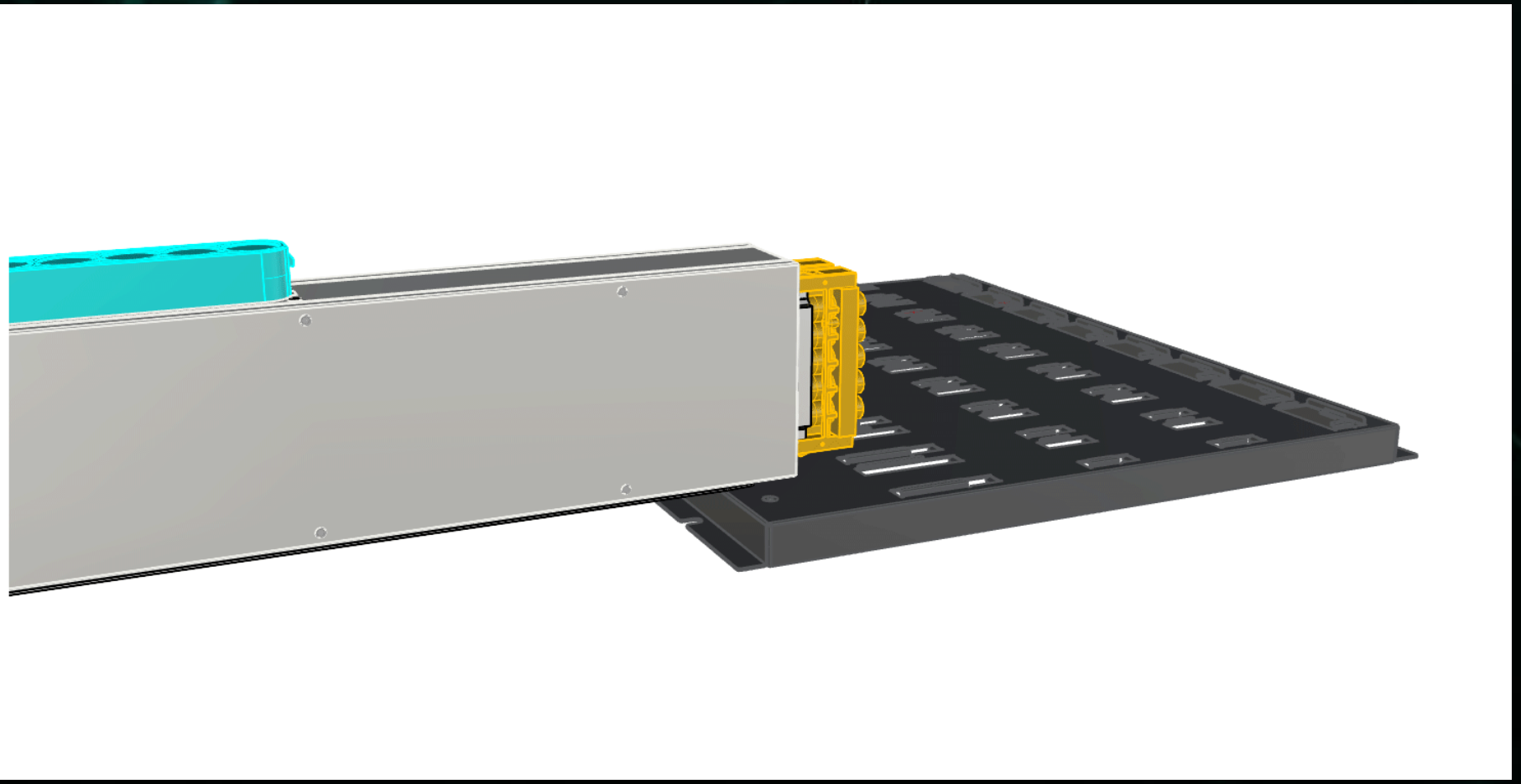
With an overall width of 600 mm, it can be installed on Rack with width >600 mm.

Feeder Box with compact width 72mm, 8 Modules can be arranged within 600 mm.

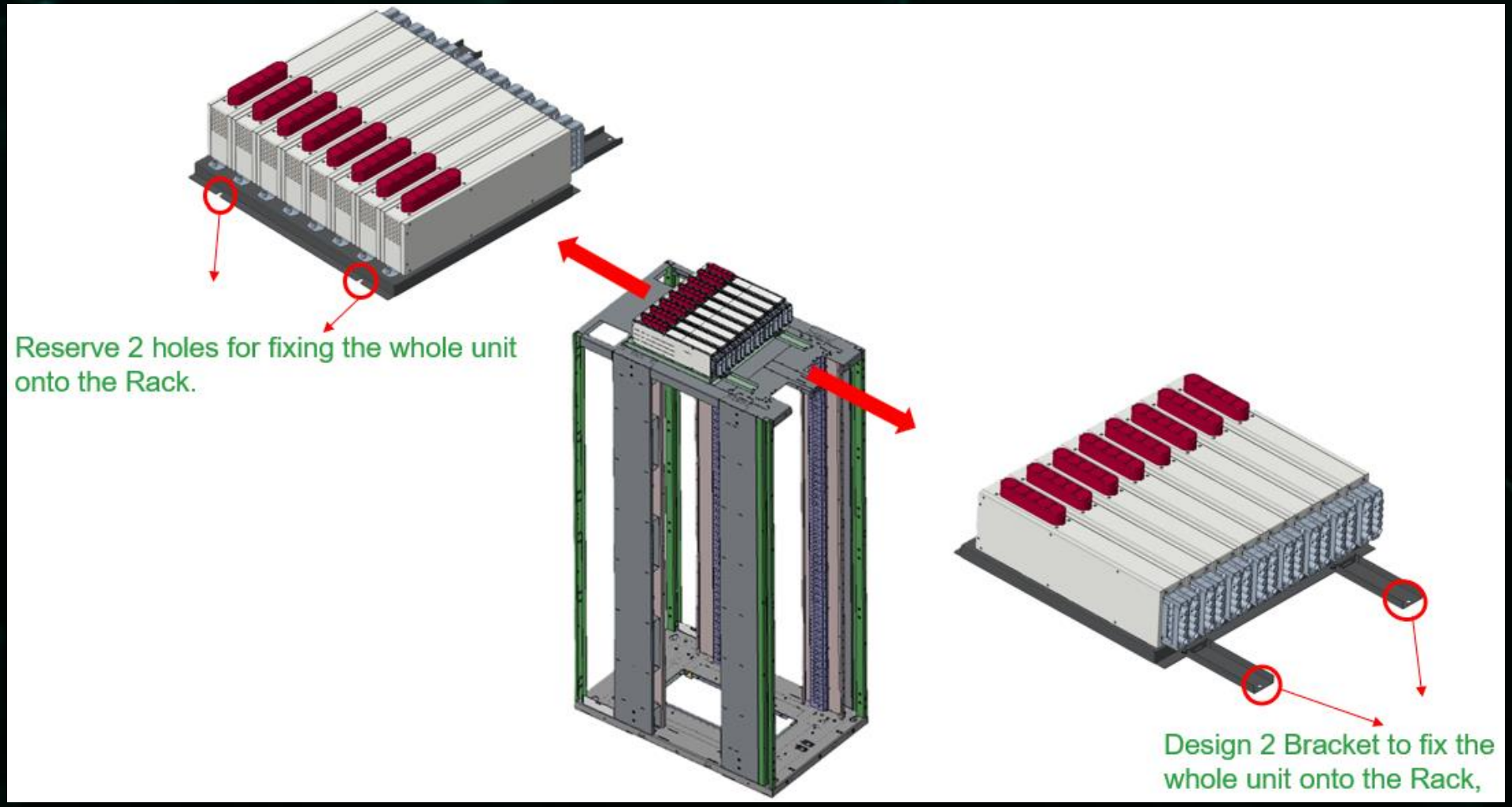


Feeder Box Modules Installation

Easy Installation, Maintenance, Scale-up (Expand to 8 Feeder Box Modules)



Horizontal AC High Power PDU installation (Customize to fix onto Rack)



Horizontal AC High Power PDU Prototype Photos

AC PDU with 4 Feeder Box Modules (could be extended to 8)



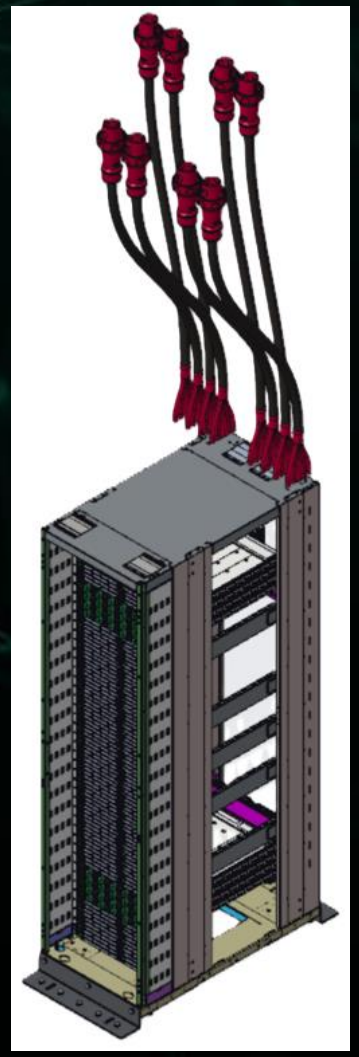
Front View



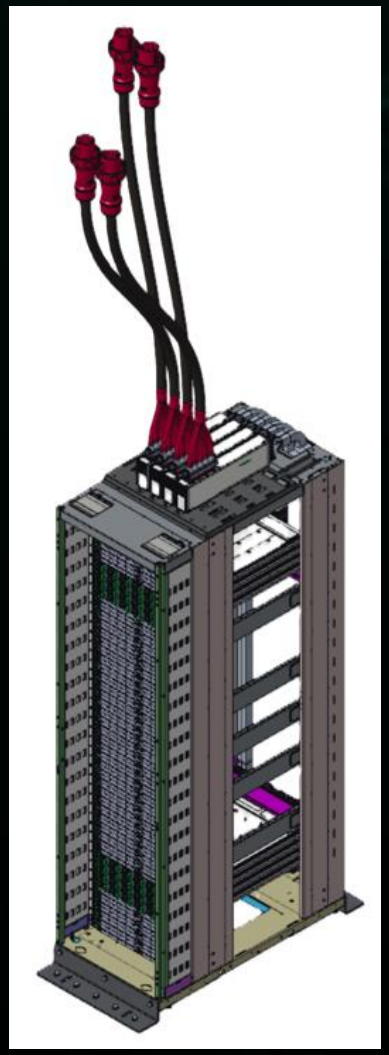
Rear View

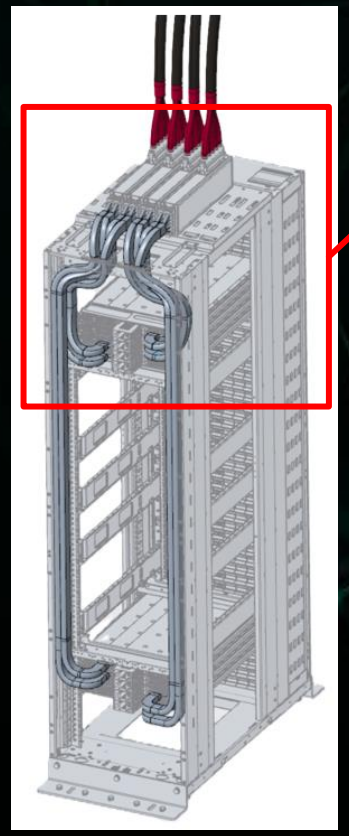
Horizontal AC High Power PDU installation illustration

Original 8 Powerwhips

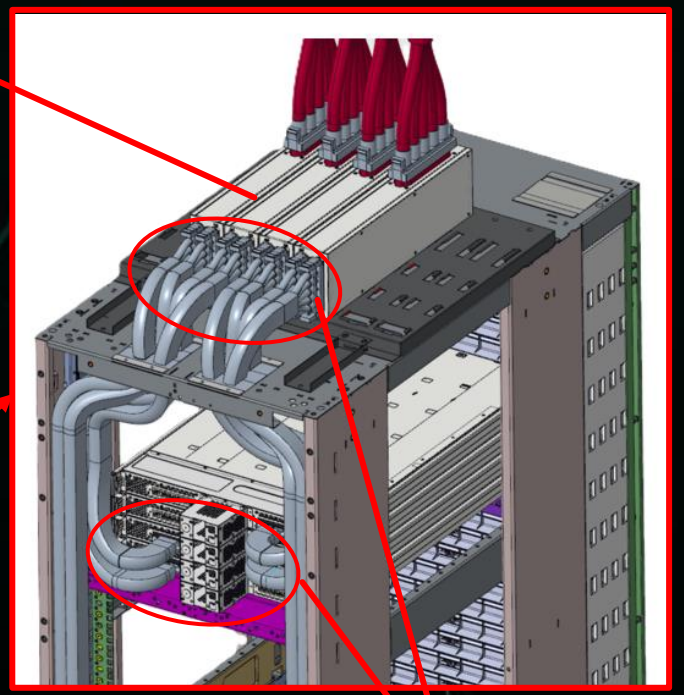


4 Powerwhips





Feeder Box Module

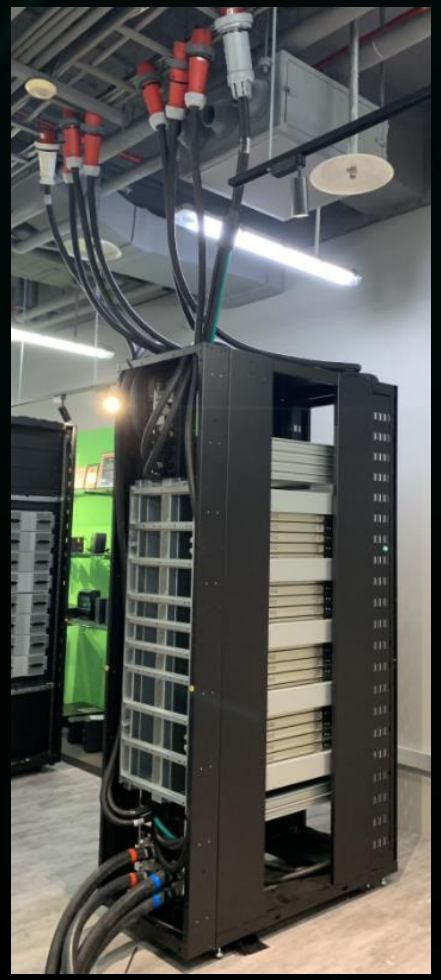


Route to Powershelves

Horizontal AC High Power PDU Prototype Photos

Original 8 Powerwhips





AC PDU with 4 Powerwhips



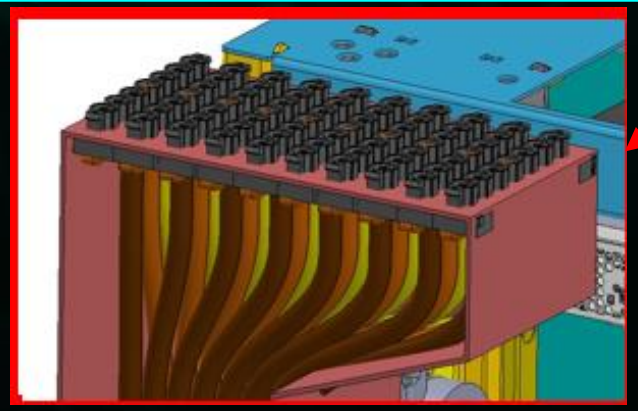


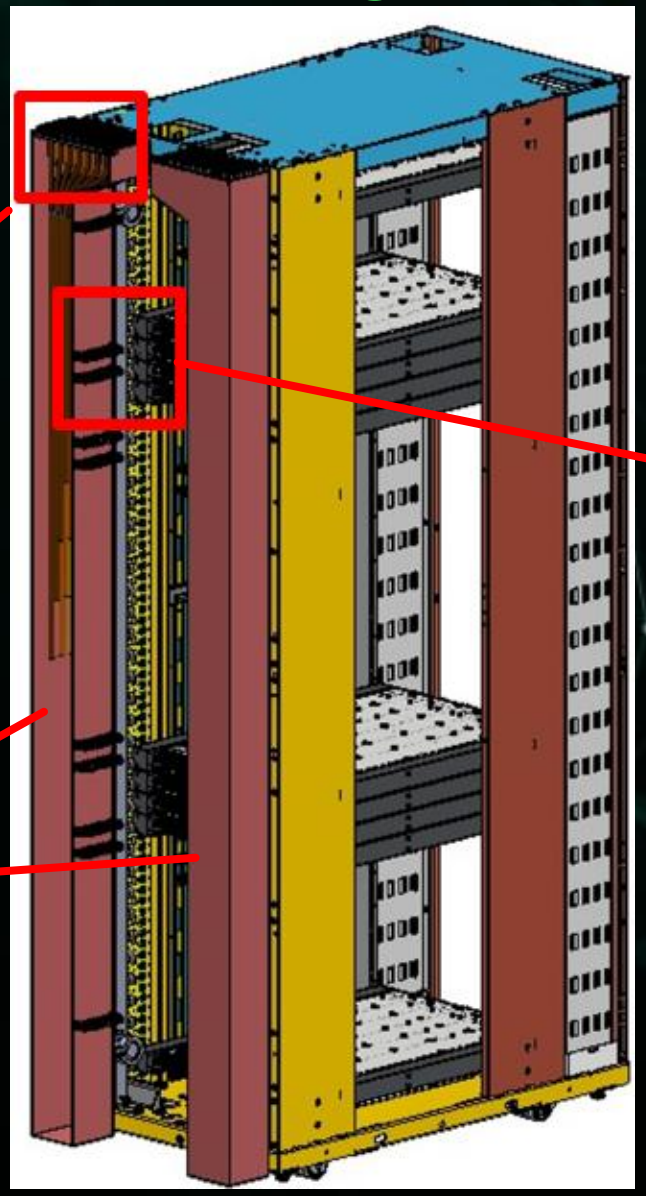
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Proposal # 2
Vertical High Power PDU hang on Rear of Rack

Proposal #2 Vertical AC High Power PDU hang on Rear of Rack

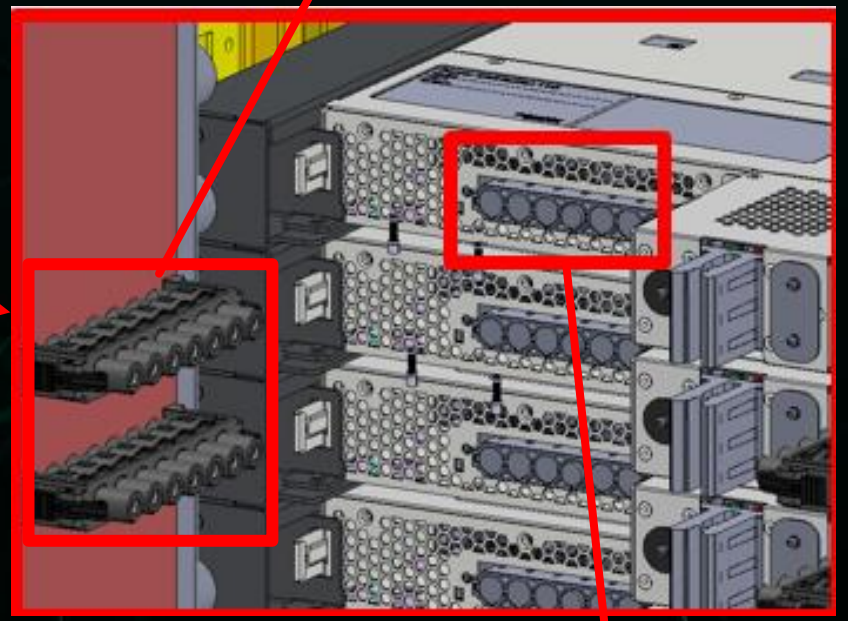
AC Input Connector of Vertical AC PDU for Powerwhips connection

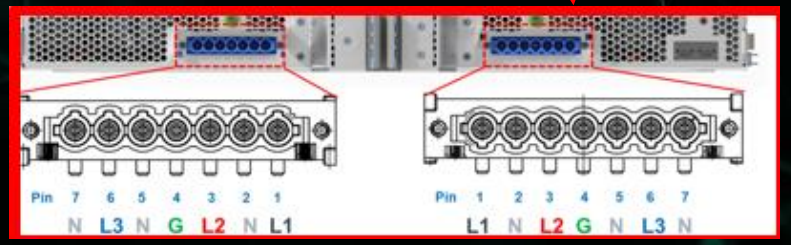




Vertical AC PDUs

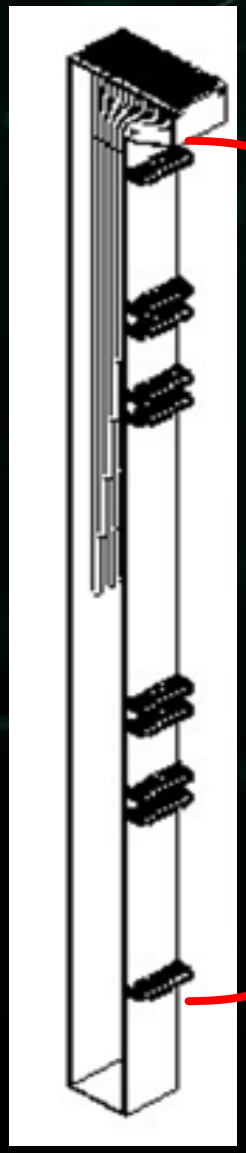
AC Output Connector of AC PDU



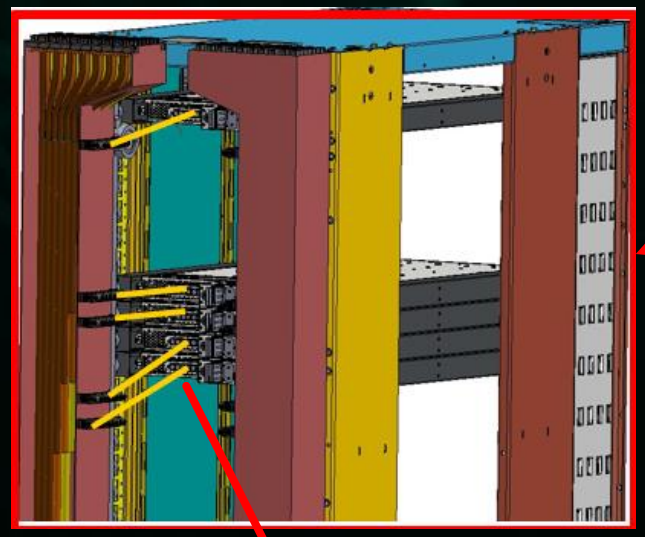


AC Input Connector of Powershelf

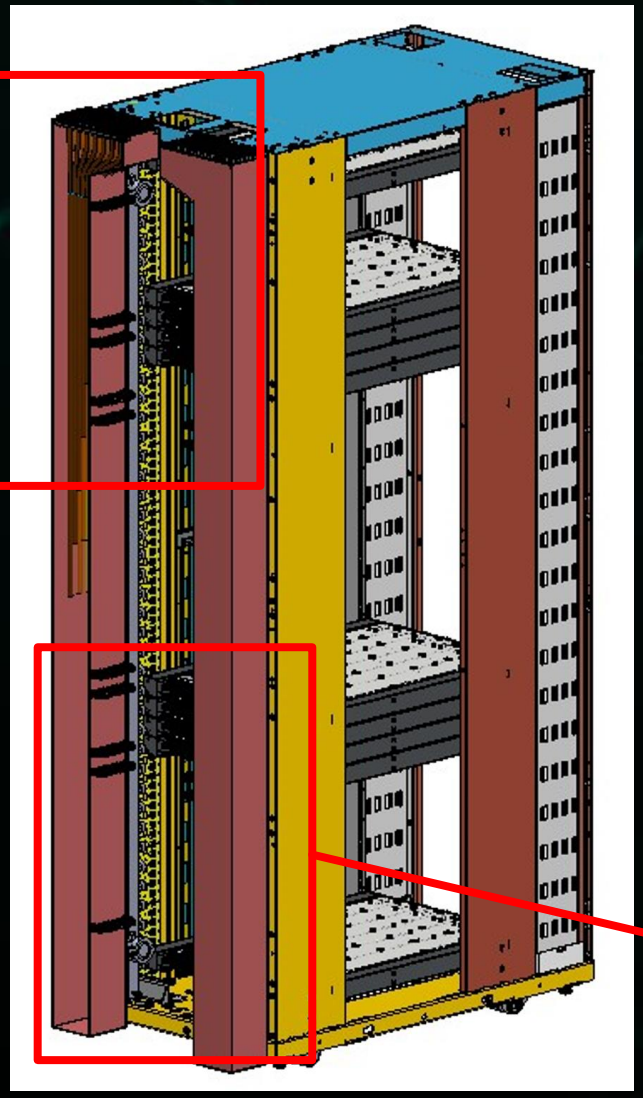
Proposal # 2 Vertical AC High Power PDU hang on Rear of Rack



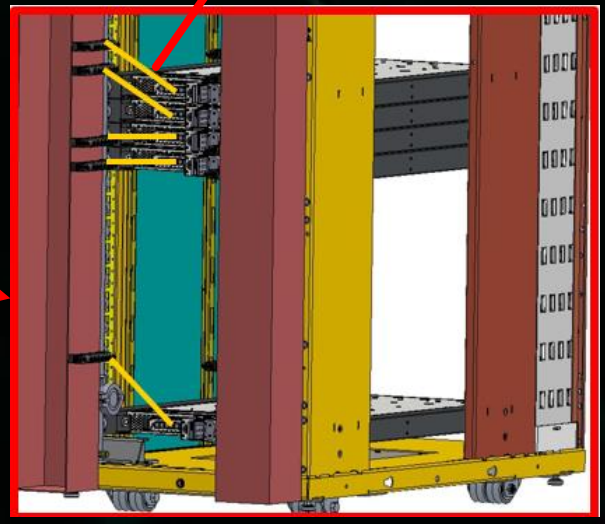
AC PDU Output Connectors



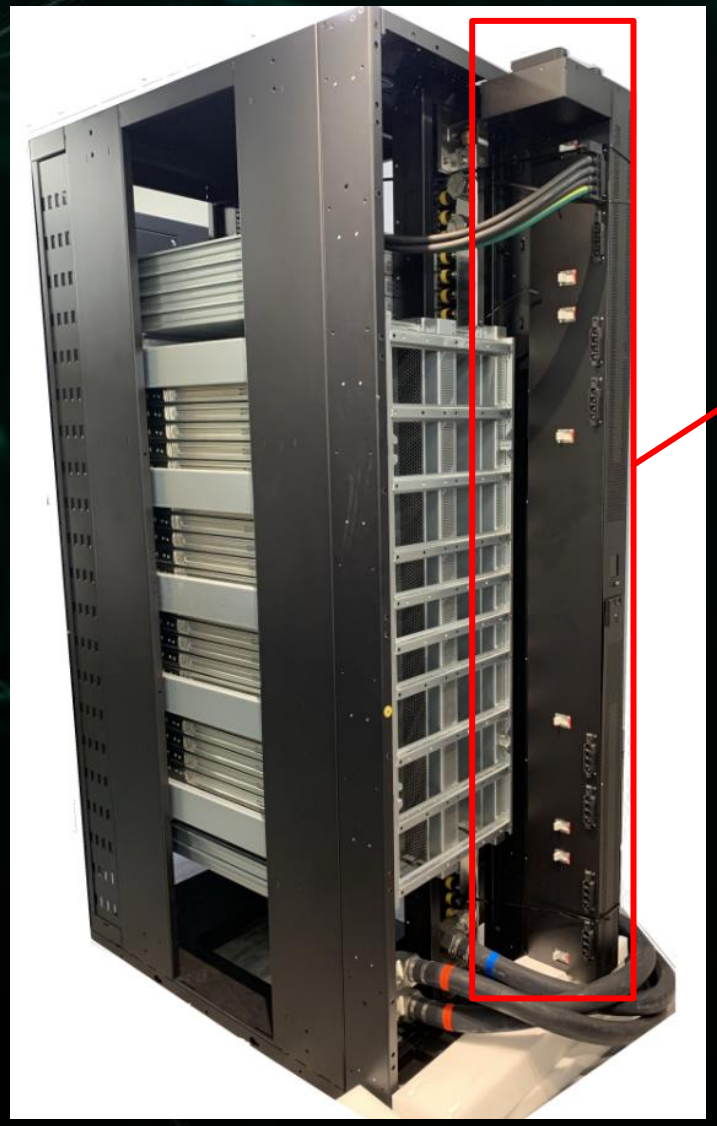
AC PDU Output Connector Connect to Powershelf Input connector



AC PDU Output Connector Connect to Powershelf Input connector

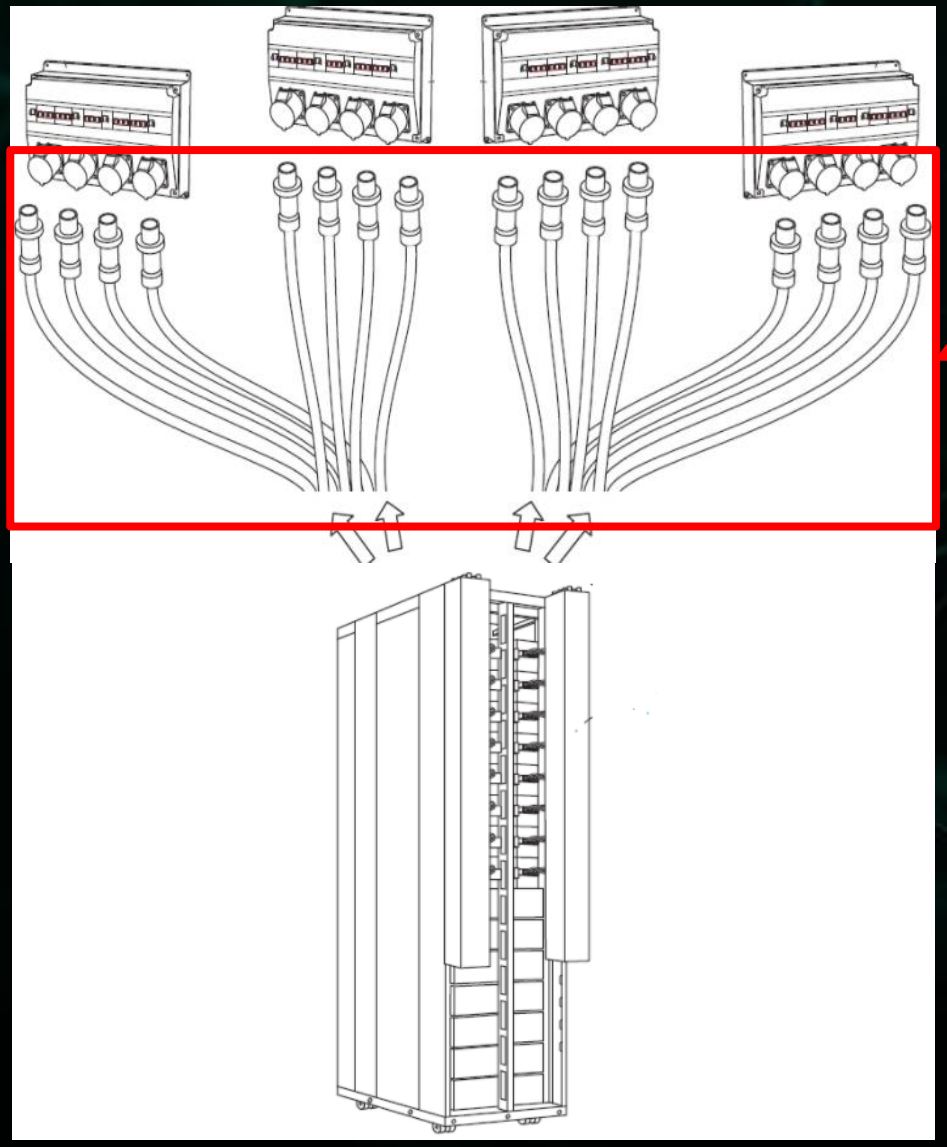


Vertical AC High Power PDU Prototype Photo





Powerwhips connect to Top - Off box



AC Powerwhips connect to Vertical AC PDU to Top-Off Box

Call To Action

- Problem to solve
  1. Consult with AI manufacture or Data Center to finalize the specification.
  2. Total Ownership Cost Evaluation

- Timeline for contribution availability
  1. Provide Prototype for Filed testing and collect feedback for improvement Q1 2027

- Timeline for product/facility availability
  Schedule to production in Q3 2027

Thank You!

